motive engineer to readjust the lubricator feeds.

The cylinder walls on a 28 x 30 in. engine have an area of about 6,000 square inches and it would require approximately 150¼ drops of oil to restore the oil film on this area, which could not be accomplished with a device that was set to supply only the necessary amount of lubricant to meet perfect running conditions. On locomotives it is considered good practice to either start the lubricator at the ordinary rate of feed some minutes before the engine leaves the initial terminal, or feed fast for a short time after leaving. The reason for this practice is simply to establish an initial oil film that had been destroyed on account of the engine being idle at the terminal and this same condition takes place when the engine is delayed for any length of time on the line of road.

On heavy freight hauls the temperature of the cylinders becomes very high on account of the long cut-off, and although the surface rubbed over per minute may be low, the temperature of the oil film will be high, and it is the temperature that reduces the life of the oil film rather than the number of times it is rubbed over, and the amount of oil required will have a closer relation to temperature than to area rubbed over when the temperature is due to a foreign agent.

The oil film is also often destroyed by the admission of front end gases to the cylinders when they are at high temperatures, and it must again be built up by the admission of a little excess oil, because the regular feed only supplies the necessary amount to replace the regular wear of the film, and cannot be expected to build up a new film. The lubricator must be equipped so its capacity can readily and quickly be adjusted to meet requirements.

Force feed lubricators have their drive rod connected by two methods: to the Walschaert link, giving it a uniform movement, or to the valve cross-head, giving it a variable movement corresponding to the cut-off. In some cases it has been connected to the top of the Walschaert combination lever, but at short cut-off this point becomes almost a stationary fulcum point, so that very little movement is imparted to the lubricator lever. The method of connecting it to the valve cross-head is undoubtedly the best, because even at the shortest cut-off it always has the lap and lead movement, but this leads to abuses, as engineers will then run with longer cut-off so as to increase the oil feed.

Some disadvantages to be noticed with the hydrostatic lubricator are that on account of the height of modern cylinders and valves and the limited height of the cab it is difficult to get the necessary slope for the oil pipes, and because of congestion in the cab and limited height it is difficult to place the lubricator so that it can be operated and filled efficiently. The increased length of the oil pipes on modern power has also increased the frictional resistance of the steam flow and thereby decreased somewhat the volume flowing through them.

With respect to the question of separate oil feed to the cylinders, it is evident that such is an improvement in cylinder lubrication. It has been rendered necessary due to the increased length of the oil pipes to the valves greatly increasing the friction and diminishing the flow of steam through them and thereby making another set of pipes necessary.

Among the replies received to the questionnaire sent out was one from the London Midland and Scottish Railway Company, stating that this railway has 513 locomotives equipped with mechanical lubricators. They have no locomotives equipped with either the two-feed or the four-feed lubricators for oiling valves and cylinders, but a large number of their engines are fitted with eight-feed mechanical lubricators, and they also have three-cylinder compound passenger engines fitted with twelve-feed mechanical lubricators. In regard to the type used, they state that the mechanical lubricator used is of their own design, each oil supply being fed by its own pump unit. Regarding tests conducted between mechanical force feed and hydrostatic lubricators, they state that they have fitted some engines with hydrostatic lubricators, but prefer the mechanical lubricator.

To the sixth question, how they compare in efficiency and what were the comparative results in regard to consumption of oil, the reply is as follows: "From our experience, the mechanical lubricator has proved very efficient and lends itself to adjustment and a constant feed, which permits of the most economical working conditions. The adjustment of the oil feed is not under the control of the enginemen. The hydrostatic lubricator adjustment is under the control of the enginemen and, being very sensitive, it is difficult to obtain economical results."

In answer to the seventh question they state that "the condition of the working surfaces of the cylinders and packing rings with mechanical lubrication is quite satisfactory, no undue wear being noticeable." As to their experience in regard to maintenance, they reply that it is very rarely necessary to interfere with the mechanical lubricator between the shopping periods of a locomotive. Also, that they have no particular data of mileage per lubricator failure, as such failures are very rare.

The talk on lubrication of valves and cylinders so far has had to do with valve oil and of different methods of supplying it to valves and cylinders, but at least one large system has secured some very good results from the use of day graphite for lubricating valves and cylinders. This graphite is prepared in the form of solidified cakes and is fed by what is known as a pendulum graphite lubricator. The magazine containing the graphite is situated on top of the boiler and back of the smoke-stack. The natural side motion of the locomotive while running causes the

				Used Pts. Valve Oil		Miles Per Pint Valve Oil			
Engine	Train	No. cars	Miles	Valves and Cylinders	Stoker Feed Water Air Pump	Car Oil	Valves and Cylinders	Stoker Feed Water Air Pump	Car Eng. Compl.	Mad. Kipp Lubricator 13/16 Cut-off
6022	14	8	289	4	3/8	8½	72	771	34	2 nicks
6022	21	9	233	2 3/8	..	6	65	...	39	3 nicks
6022	20	8	233	1 1/4	..	6	40	...	39	3 nicks
6022	17	9	289	7 1/4	..	5	40	...	57	3 nicks
6022	110-18	8	179	2 1/8	..	4	61	...	45	2 nicks
6022	9	10	179	2	..	4	72	...	45	2 nicks
6033	112-16	8	289	3	..	6	96	...	48	2 nicks
6033	15	8	289	4	..	6	72	...	48	2 nicks
6022	110-18	12	179	3 1/8	..	4	46	...	48	2 nicks
6022	9-7	13	179	3 3/4	..	4	48	...	48	2 nicks

The hydrostatic lubricator in the cal. was set to feed 1 drop per minute.

pendulum to swing over the cakes of graphite and dusts or scrapes off a small amount of graphite at each swing. The graphite scraped off passes through a pipe into the superheater header and extends to the valves and cylinders.

You will observe that this method of lubrication has the same constant effect as the mechanical lubricator, as it only feeds while the engine is in motion. It is the opinion of those who have had experience with this method of lubrication that the graphite passing through the superheat units has the effect of preventing leaks developing in same. It is noted that engines equipped with the graphite lubricator did not give as much trouble with leaky units as other engines operat-

Mechanical Division Meeting of A. R. A.

Abstracts of Some of the Reports Presented at Chicago Meeting

The annual meeting of the Mechanical Division, American Railway Association, was held at the Hotel Drake, Chicago, Ill., June 16, 17 and 18. J. J. Tatum, superintendent, car department, Baltimore & Ohio Railroad, chairman of the Division, presided.

In accordance with the decision to hold meetings with exhibits only every other year, the annual meeting this year was a strictly business meeting without exhibits of machinery and devices. However, a small exhibit of equipment was arranged by the railroads in the yards of the Illinois Central. This consisted of box cars built to the design of the Committee on Car Construction, and including the single sheathed car adopted as recommended practice, the double sheathed wooden sheathed box car now being recommended for adoption as recommended practice by the same committee, and the all steel box car built to design prepared by the Committee two years ago.

In addition to these cars several modern locomotives were exhibited by their railroad owners.

R. H. Aishton, president, American Railway Association addressed the opening session, following which the reports of the various committees and individual papers were dealt with.

The membership of the Division at the present time includes 208 railways, representing 399 memberships in the American Railway Association, and in addition thereto, 190 railroads, associate members of the American Railway Association. These railroads, members and associate members of the American Railway Association, have appointed 999 representatives in the Mechanical Division. In addition there are 1,065 affiliated members and 142 life members in the Division.

In this and the following pages is presented abstracts from some of the Committee reports.

Locomotive Design and Construction

This report consists of a series of practically independent reports on the following subjects:

Comparative Merits of Hydrostatic and Force Feed Lubrication for Locomotive Cylinders and Steam Chests and Best Methods of Application;

Standardization of Taps and Dies Used by Railroads, Bolt and Screw Thread Standardization;

Definition of an Engine Failure;

Standardization of Water Columns;

Removable Hand Rail Columns;

Locomotive Development.

This last was divided into two parts: The Steam Locomotive and the Internal Combustion Locomotive.

Hydrostatic and Force Feed Lubrication

That there has been considerable interest manifested in the process of force feed lubrication of locomotive valves and cylinders is evidenced by the number of roads that have force feed lubricators in use, either experimentally or established as a practical system. Inquiries sent to the manufacturers of locomotive force feed lubricators requesting a statement as to the number of lubricators in service or on order as of March 1, 1925, developed that there are now in service, or on order, force feed lubricators of the various types, as follows:

Nathan Mfg. Co.	386
United States Metallic Packing Co.) (McCord)	694
Formerly Loco. Lubricator Co.) (Schlacks)	1308
Edna Brass Mfg. Co.	36
Madison-Kipp Corp. (Information not available)	
	2,424

These lubricators are distributed over quite a wide range including 76 railroads and 13 logging or commercial plants owning locomotives.

The expressions received from the various railroads using force feed lubricators indicate that this method of lubrication is exciting considerable interest, and is establishing a favorable impression. Only in a very limited degree was any expression adverse to force feed lubrication received.

There is a total of seven roads that co-operated with your Committee in the pursuit of this subject and have furnished the results of their observations. The reports from these seven roads embrace the comparative performance of hydrostatic versus force feed lubrication on a total of 22 locomotives. With the exception of four, these locomotives were equipped with hydrostatic oil delivery to one side and force delivery to the opposite side.

In preparing the table showing the relative results obtained from the two systems of lubrication, the record as obtained in passenger and freight and locomotive service has been separated according to the class of service, following which a combination of the results, including both passenger and freight service, is shown. There were 14 passenger locomotives under observation, during which time 38 valve rings in the hydrostatically lubricated positions, and 45 in the force feed lubricated positions were removed on account of being worn, broken or down. From these valve rings an average service of 13,471 miles was obtained with hydrostatic lubrication, and 11,749 miles with force feed lubrication. From these same locomotives there was a total of 72 and 104 cylinder packing rings removed from the hydrostatic and force feed positions respectively. The average mileage per cylinder ring was 7,600 for the hydrostatic, and 6,550 for the force feed. A representative number of rings removed were measured for radial wear at five equally spaced points around the ring from the ends, and from the mileage performance record of these rings the average rate of wear per 10,000 miles has been determined, which, for the hydrostatic lubrication, was .027 inches; force feed lubrication .029 inches, the rate of wear was practically equal. Comparing the mileage made per pint of oil there is a marked difference favorable to force feed lubrication since an average of 50.6 miles per pint of oil was secured as against 30.5 miles for the hydrostatic. Attention is called, however, to the wide range in the oil consumption figures as indicated by the maximum and minimum results. With hydrostatic lubrication the maximum and minimum mileage per pint was 49.1 to 15.9 miles respectively, while with the force feed lubricator the range was from 101.8 to 21.4 miles respectively.

In freight service there was a total of eight locomotives under observation. The performance of the valve packing rings was not sufficiently complete to enter into the com-

was obtained from the force feed system, while a better mileage performance per pint of valve oil was obtained from the hydrostatic system. While there may be a slight difference favorable to force feed lubrication as against hydrostatic system, the difference is not striking or particularly conclusive.

As previously mentioned, some very enthusiastic expressions were received from roads using or experimenting with force feed lubrication, but in the consideration and analysis of the concrete data furnished your Committee covering the performance record obtained from the two systems of lubrication the results obtained from force feed lubrication do not stand out in strong contrast with those obtained from the hydrostatic. There may be those who are strong advocates of either system of lubrication who may feel disposed to challenge the results presented, and the position they assume may be the result of observations of individual comparisons where the differences were strikingly established, but considering 22 locomotives operating on seven different roads, the results obtained do not strongly support one system of lubrication in preference to another.

There is unquestionably a strong tendency developing towards force feed lubrication, and it is the opinion of your Committee that this tendency is in the line of progress. It may not be amiss in this connection to mention a few of these more pertinent items that should be given consideration in the installation of force feed lubricators:

The lubricator should be firmly supported by being bracketed, preferably to the cylinder casting, and should be located so it will be readily accessible for filling and for such attention as to adjustment of feeds as may be necessary. In all installations the piping leading from the lubricators to the point of delivery should be as short as possible and secured to some rigid portion of the locomotive, such as following along the frame construction or bracketed to the cylinder casting. Encasing the tubing in electric loom provides some protection against changes in atmospheric temperature.

In order that the oil feed may be in direct proportion to the piston travel regardless of the position of the reverse bar it is recommended that the lubricator be actuated by connection with the valve motion link or other positive travel location.

It might be well to recognize certain features of the force feed lubricator that are considered desirable. Suitable means for controlling the temperature of the oil in the lubricator should be provided. If the temperature of the oil is too low, delivery will be retarded. If too much steam for heating the oil reservoir is turned on, the oil will eventually boil over with resultant waste. Provision should be made for individual feed adjustment, since in some installations where oil is delivered to both valves and cylinders it is desirable to have some method of adjustment of the oil feed so that each individual delivery can be adjusted to the amount of oil required for that particular position.

After careful consideration of all the information collected in the study of the subject of hydrostatic and force feed lubrication of locomotive valves and cylinders it is the thought of your Committee that this is a subject for which no general rule or recommendation can consistently apply. The conclusion therefore reached is that on account of local influence or existing conditions on different roads the solution as to whether hydrostatic or force lubrication will meet conditions to the better advancement of economy and service will have to be worked out by individual roads having the subject under consideration.

Standardization of Taps and Dies and Bolt and Screw Threads

This Committee has no comment or criticism to offer with respect to the standards, already adopted by the Association, except to suggest that in view of the disposition of the railroads and the Association to adopt screw threads which do not conform to the Sellers or U. S. Standard tables, particularly for special purposes, such as boiler work, pipe work, injector couplings, etc., the statement on page L-27 of the Manual, to the effect that "The Sellers or Franklin Institute System of screw threads is the standard of the Association" should be qualified so as to avoid conflict between this statement and other standards and recommended practices which have been, or may be adopted.

The Committee feels that in the case of thin castle nuts, whose use, insofar as locomotives are concerned, is confined almost exclusively to crank pins, knuckle pins, crosshead pins, piston rods, etc., the form and number of threads per inch in the nuts should be governed by the threading of the parts to which they are applied.

The Committee also feels that the form of threads to be used for lubricator fittings should be definitely specified.

Attention has been called by the Committee on Locomotive and Car Lighting to the fact that the Association has no standards for bolts or machine screws smaller than ¼-in. diameter, which is the smallest size included in the Sellers or U. S. Standard tables. This Committee has therefore undertaken the work of ascertaining the present practices of the various roads with regard to sizes and threading of machine screws.

Information has been received from twenty roads. Fifteen report having no standards of their own, but are either following, or recommend following, the A. S. M. E. standards. Three have adopted as standard a certain number of sizes which are included in the A. S. M. E. standards, but have not adopted the entire A. S. M. E. table. Two roads report having standards of their own which do not agree with the sizes and threading found in the A. S. M. E. tables.

In view of the predominance of the A. S. M. E. sizes, it seems proper that the sizes and threads that are adopted for machine screws should be selected from the A. S. M. E. tables. Attention is called to the fact that the A. S. M. E. tables include screws up to, and including .45 inches diameter, thereby including some sizes which are practically the same, as regards diameter, as sizes included in the U. S. Standard table, but in all cases the A. S. M. E. threading differs from the threading of corresponding sizes shown in the U. S. S. table. We therefore recommend adoption of certain A. S. M. E. sizes of machine screws as shown in Table 1.

The A. S. M. E. standards include five styles of heads for machine screws, some of which have obviously been designed to present an elegant appearance. For locomotive use, the Committee recommends consideration of only three styles of heads, viz.:

The Flat or countersunk head, Round or button head, Flat fillister head.

The Flat fillister head, last mentioned, may be either round or hexagonal.

The Committee has endeavored to collect information as to the practices followed by different railroads in the matter of number of threads per inch, form or profile of threads and tapers applying to the following screw threaded parts of locomotives:

Staybolts, Crosshead pins, Crank pins, Knuckle joint pins, piston rods, Washout and arch tube plugs.

Information with regard to staybolt threading was

Mechanical Appliances for Lubricating Locomotives

Continuation of Reports to Traveling Engineers' Convention

There are varying conditions under which locomotives are operated, such as the difference existing in the locomotive with properly fitted pistons and rings and one which has nearly completed its mileage; also, the difference existing in lubricating the saturated and the superheated locomotives, as the superheated locomotive with a higher degree of heat breaks down the oil film so much more rapidly; again, the difference existing in operating an engine in good water territory as compared with the alkali waters of the West and Southwest; also, the locomotive of medium steam pressure as compared with locomotives of higher pressure. Again, we have locomotives operating in through service, which require different handling of lubrication features as compared with engines in local service.

These points are mentioned in order to draw attention to the difficulty in recommending a definite amount of oil to be used per mile. In arranging for the proper lubrication of an engine, the above mentioned different features must necessarily be taken into consideration. Before the advent of the superheated engines, the question of lubricating valves and cylinders did not present a very serious problem, but the introduction of the superheater with its high degree of superheated steam and fast and heavy service made it necessary to provide a means of furnishing a constant and uniform supply of oil to the valves and cylinders.

Until recently the hydrostatic lubricator of various designs had been the only medium used for applying oil, but with the advent of larger locomotives it has been felt that a device should be produced which would fulfill these functions more efficiently. With this end in view, the mechanical or force feed lubricators of various designs have been placed on the market, and have been applied to a considerable extent. The committee sent out a questionnaire to the membership with a view of securing information as to the degree of success secured from the application of the force feed lubricators; also, how their operation compares with the hydrostatic lubricators. The replies received would indicate that while a great number of those lubricators have been applied, they have not been in service long enough to secure definite data as to their efficiency as compared with the hydrostatic lubricators.

Each of the two types of lubricators has its separate merits and defects. For instance, by applying the hydrostatic lubricators to the large modern engines, it has been felt by some that due to the great length of the oil delivery pipes serious difficulties were being introduced in delivering the oil successfully to the valves and cylinders.

Also, in long passenger runs great difficulty has arisen through engineers covering the first divisions using the greater portion of the oil supply and, in many cases, leaving the man covering the last lap of the run short of oil. This feature would by the mechanical or force feed lubricator be taken out of the control of the engineer, thus insuring a continuous and constant supply of oil covering the entire trip. The amount to be supplied can be regulated to cover the requirements demanded by the class of service in which the engine is working.

Not being able to secure any definite data as regards to the comparison of the hydrostatic and mechanical lubricators, the chairman of this committee will give his personal observation. We have applied mechanical force feed lubricators to our mountain type locomotives operating in heavy fast passenger service and have secured very satisfactory results. Certain defects were encountered, but with the experience we have gained these are readily overcome. The chief defect was choke valves leaking, which allowed steam to back up in the oil pipes and condense in the oil chamber. This condensation settled to the bottom and in some cases was of such a volume as to float the oil above the intake of the pump. This defect can be easily detected and remedied by the simple practice of opening the drain valve at the bottom of the lubricator and if water appears, this is an indication that a leaky choke valve exists and requires attention. Our staff has now been instructed and trained to watch this feature. The result is that this defect has now been practically eliminated. There are also instances where the lubricator apparently did not feed on one side. The cause of this was found to be choke fittings corroded and seized.

I would like to call attention to a point which came under my personal observation. We had two 34% engines of the Pacific type in heavy local passenger service. One of the engines was apparently not getting sufficient oil in one side. A test showed that the lubricator was delivering the oil all right. Further examination disclosed that the piston rings on this side were worn, causing a blow, but it was not sufficient to be detected out of the stack. The rings were changed and no further trouble was experienced in regard to the lubricator. I wish to bring out this fact that the oil supplied by the mechanical lubricator had been adjusted to a nicety and was sufficient so long as the rings were tight, but as soon as the rings started to blow there was not then sufficient lubrication, and I became convinced that the properly adjusted mechanical lubricator would immediately show up the defect when the engine started to blow. I do not doubt but that had this engine been equipped with a hydrostatic lubricator the engineer would simply have adjusted the lubricator to get more oil to that side and the blow would have continued with the consequent loss of fuel until sufficient steam would be escaping to be detected through the stack.

Some authorities claim that a lubricator should feed an amount of oil corresponding to the speed of the locomotive. This is not good reasoning, because the volume of oil required for lubrication is not always in proportion to the velocity, or to the area rubbed over, and especially is this so when the temperature of the rubbing surfaces is increased by some other agent than friction, which greatly reduces the viscosity of the lubricant and destroys the oil film in a very short time.

On this account the volume of cylinder lubrication would be regulated more by the cylinder temperature than by the area rubbed over. Assuming, then, that on account of occasional high water or bad water the oil film on the cylinder walls has been destroyed, how is this lubricating film to be restored if the volume of lubricant cannot be immediately increased? I might say that this film can be restored with the mechanical lubricator by lengthening the cut-off, which most generally would come in line when a condition of this kind arises. With a hydrostatic lubricator it would be necessary for the loco-